॥वसुधैव कुटुम्बकम्॥
SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment/ Case Study No.: 1

1. Experiment Title: Demonstrate use of inline, internal, and external JavaScript. Use console methods such as console.log(), console.error(), console.table(), and console.trace(). Create a simple webpage that logs user environment info and shows a welcome message.

2. Software/Tools Required:

- Visual Studio Code (VS Code) / Notepad++
- Google Chrome / Mozilla Firefox
- JavaScript (ES6)
- HTML5
- Windows/Linux Operating System

3. Experiment Program Code:

1. index.html:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>JavaScript Practical 1</title>

  <!-- Internal JavaScript -->
  <script>

    function showMessage() {
      alert("Hello Lavanya Tiwade! This is Internal JavaScript.");
    }

    console.log("===== JavaScript Practical 1 =====");
    console.log("Student Name: Lavanya Tiwade");
    console.log("PRN: YOUR_PRN_NUMBER");

    console.error("This is a sample error message.");
```

```
    console.table([
      {
        Name: "Lavanya Tiwade",
        PRN: "YOUR_PRN_NUMBER",
        Course: "B.Tech CSE"
      }
    ]);

    function traceExample() {
      console.trace("Trace Example");
    }

    window.addEventListener("load", function () {

      document.getElementById("greeting").innerHTML =
        "<h2>Welcome Lavanya Tiwade! 👋</h2>";

      console.log("Browser:", navigator.userAgent);
      console.log("Platform:", navigator.platform);
      console.log("Language:", navigator.language);
      console.log("Online:", navigator.onLine);

      traceExample();
    });

  </script>

  <!-- External JavaScript -->
  <script src="script.js"></script>

</head>

<body>

  <h1>JavaScript Practical 1</h1>

  <p><strong>Name:</strong> Lavanya Tiwade</p>

  <p><strong>PRN:</strong> YOUR_PRN_NUMBER</p>

  <div id="greeting"></div>

  <p>This page demonstrates Inline, Internal and External JavaScript.</p>

  <!-- Inline JavaScript -->
  <button onclick="alert('Hello Lavanya Tiwade! Welcome to JavaScript Practical 1.')">
    Inline JavaScript
```

```
    </button>

    <br><br>

    <!-- Internal JavaScript -->
    <button onclick="showMessage()">
       Internal JavaScript
    </button>

    <br><br>

    <!-- External JavaScript -->
    <button onclick="externalMessage()">
       External JavaScript
    </button>

</body>

</html>
```

2. script.js:

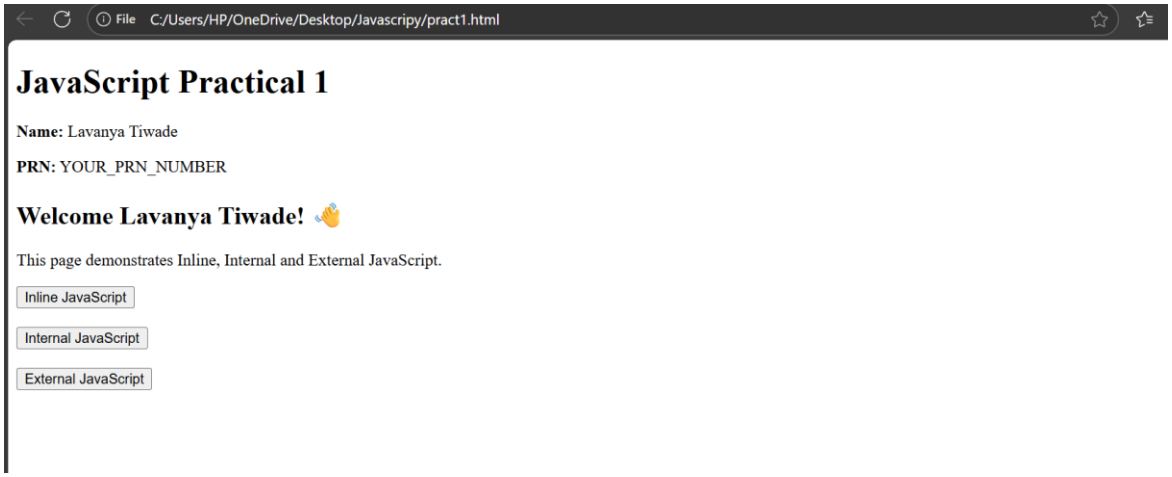
```
function externalMessage() {
    alert("Hello Lavanya Tiwade! This is External JavaScript.");

    console.log("External JavaScript Executed");

    console.table([
       {
          Browser: navigator.userAgent,
          Platform: navigator.platform,
          Language: navigator.language,
          Online: navigator.onLine
       }
    ]);

    console.trace("External JavaScript Trace");
}
```

4. Output:



5. Case Study Title: **Student Information and Browser Details Portal**

6. Case Study Program Code:

1. index.html:

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Student Information Portal</title>

<script>

function showInternal(){

alert("Internal JavaScript Executed");

}

function displayStudent(){

var name=document.getElementById("name").value;

document.getElementById("welcome").innerHTML="<h2>Welcome "+name+"
👋</h2>";

console.log("Student Name :",name);
```

```
console.log("Browser :",navigator.userAgent);

console.log("Platform :",navigator.platform);

console.log("Language :",navigator.language);

console.log("Online :",navigator.onLine);

console.error("Sample Error Message");

console.table([
{
Name:name,
Platform:navigator.platform,
Language:navigator.language,
Online:navigator.onLine
}
]);

console.trace("Trace Example");

}

</script>

<script src="script.js"></script>

</head>

<body>

<h1>Student Information Portal</h1>

<input type="text" id="name" placeholder="Enter Your Name">

<br><br>

<button onclick="displayStudent()">Submit</button>

<div id="welcome"></div>

<br>

<button onclick="alert('Inline JavaScript Executed')">

Inline JavaScript
```

```
</button>

<br><br>

<button onclick="showInternal()">

Internal JavaScript

</button>

<br><br>

<button onclick="externalMessage()">

External JavaScript

</button>

</body>

</html>
```

2. script.js:

```
function externalMessage(){

alert("External JavaScript Executed");

console.log("External JavaScript Loaded Successfully.");

}
```

- **Inline JavaScript:** JavaScript code is written directly inside an HTML element using an event attribute.
  `<button onclick="alert('Hello')">Click</button>`

- **Internal JavaScript:** JavaScript code is written inside the <script> tag in the HTML document.
  ```
  <script>
  console.log("Hello");
  </script>
  ```
- **External JavaScript :** JavaScript code is written in a separate .js file and linked using:
  `<script src="script.js"></script>`

- JavaScript code is written in a separate .js file and linked using: `<script src="script.js"></script>`

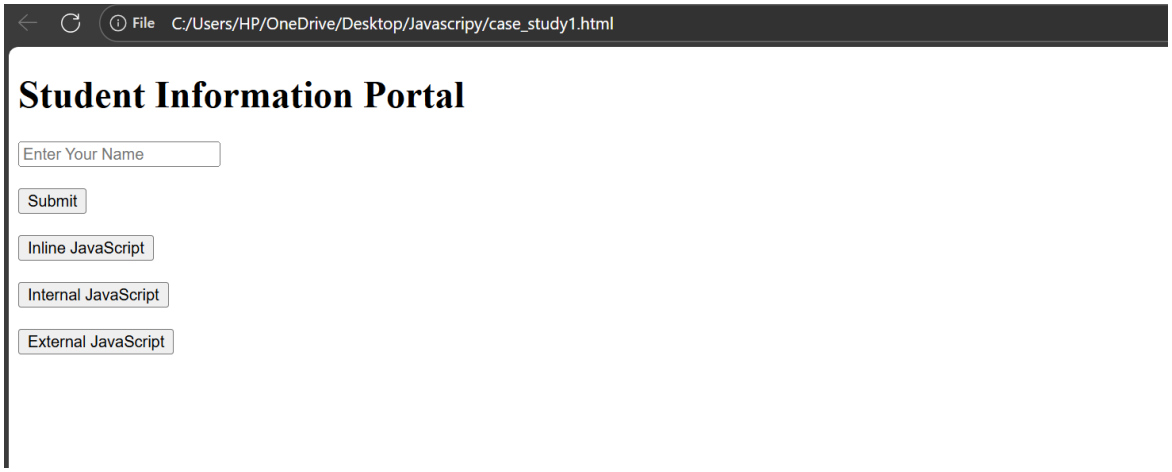
Console Methods Used console.log()
Displays normal information. console.log("Welcome");

console.error() Displays
error messages.
console.error("Sample Error");

console.table()
Displays data in table format. console.table([
{Name:"Aditi",Course:"BCA"},
{Name:"Rahul",Course:"BCA"}
]);

console.trace()
Shows the execution path (call stack).
console.trace("Trace Example");

7. Output:



8. Result/Conclusion:

The program was successfully executed. The webpage demonstrated Inline, Internal, and External JavaScript. The console methods (console.log(), console.error(), console.table(), and console.trace()) displayed the expected output, user environment information was logged successfully, and the welcome messages were shown as intended.